

Magnetism and Matter

Question1

A sample of a ferromagnetic iron in the shape of a cube of side $1.0\mu\text{ m}$ contains 8.7×10^{28} atoms per cubic metre and the magnetic dipole moment of each iron atom is $93 \times 10^{-24}\text{ Am}^2$. Then, the maximum possible magnetic dipole moment (in Am^2) of the sample is nearly

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Options:

A.

$$8.1 \times 10^{-12}$$

B.

$$8.1 \times 10^{-14}$$

C.

$$81 \times 10^{-14}$$

D.

$$81 \times 10^{-16}$$

Answer: C

Solution:

Total number of atoms in sample $N = \text{Number of atom per cubic metre} \times \text{Volume of the sample}$

$$= 8.7 \times 10^{28} \times (1 \times 10^{-6})^3$$

$$= 8.7 \times 10^{10} \text{ atoms}$$

Maximum possible dipole moment

$$\begin{aligned}M_{\max} &= N \times \text{magnetic dipole moment of} \\ &\quad \text{each atom} \\ &= 8.7 \times 10^{10} \times 9.3 \times 10^{-24} \\ &= 8.09 \times 10^{-13} = 8.1 \times 10^{-13} \\ &\approx 81 \times 10^{-14} \text{Am}^2\end{aligned}$$

Question2

A sample of paramagnetic salt contains 2×10^{24} atomic dipoles each of dipole moment $15 \times 10^{-23} \text{JT}^{-1}$. The sample is placed under homogeneous magnetic field of 0.6 T and cooled to a temperature 4.2 K . The degree of magnetic saturation achieved is 20%. Then total dipole moment of the sample for a magnetic field of 0.9 T and a temperature of 2.8 K is

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Options:

A.

$$4.5 \text{JT}^{-1}$$

B.

$$13.5 \text{JT}^{-1}$$

C.

$$0.64 \text{JT}^{-1}$$

D.

$$7 \text{JT}^{-1}$$

Answer: B

Solution:

We use Curie's law to understand how the magnetism of a paramagnetic material changes with temperature and magnetic field.

According to Curie's law, the magnetisation (M) is directly proportional to the magnetic field (B_0) and inversely proportional to the temperature (T):

$M \propto \frac{B_0}{T}$ or, $M = C \cdot \frac{B_0}{T}$ where C (Curie's constant) stays the same for the same substance.

Curie's constant can be written as: $C = \frac{MT}{B_0}$ where:

- T = temperature in Kelvin
- B_0 = external magnetic field
- M = magnetisation (total dipole moment)

Since C (Curie's constant) is the same for all measurements of the same sample, we can relate two cases as:

$$\frac{M_1 T_1}{B_{0_1}} = \frac{M_2 T_2}{B_{0_2}} \text{ So, } M_2 = M_1 \cdot \frac{T_1}{T_2} \cdot \frac{B_{0_2}}{B_{0_1}}$$

Now, let's fill in the given values:

- Number of dipoles = 2×10^{24}
- Dipole moment per dipole = $15 \times 10^{-23} \text{ JT}^{-1}$

Step 1: Find Initial Total Dipole Moment

Total dipole moment possible (if 100% aligned): $2 \times 10^{24} \times 15 \times 10^{-23} = 3 \times 10^2 \text{ JT}^{-1}$ But, at first, only 20% saturation is achieved: $M_1 = 0.2 \times 3 \times 10^2 = 60 \text{ JT}^{-1}$

Step 2: Use Curie's Law for New Conditions

Magnetic field changes from $B_{0_1} = 0.6 \text{ T}$ to $B_{0_2} = 0.9 \text{ T}$ and temperature from $T_1 = 4.2 \text{ K}$ to $T_2 = 2.8 \text{ K}$.

Now, use the formula: $M_2 = M_1 \times \frac{T_1}{T_2} \times \frac{B_{0_2}}{B_{0_1}}$

Substitute the numbers in: $M_2 = 60 \times \frac{4.2}{2.8} \times \frac{0.9}{0.6}$

Calculate step by step:

- $\frac{4.2}{2.8} = 1.5$
- $\frac{0.9}{0.6} = 1.5$

So, $M_2 = 60 \times 1.5 \times 1.5 = 135 \text{ JT}^{-1}$

Question3

The work done in rotating a bar magnet which is initially in the direction of a uniform magnetic field through 45° is W . The additional work to be done to rotate the magnet further through 15° is

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Options:

A.

$$\frac{W}{\sqrt{2}}$$

B.

$$\frac{W}{2}$$

C.

$$W\sqrt{2}$$

D.

$$2W$$

Answer: A

Solution:

$$W = MB(\cos \theta_1 - \cos \theta_2)$$

$$\text{Initially, } \theta_1 = 0^\circ, \theta_2 = 45^\circ$$

$$\begin{aligned}\therefore W_1 &= MB(\cos 0 - \cos 45^\circ) \\ &= MB\left(1 - \frac{1}{\sqrt{2}}\right) \\ &= \left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)MB\end{aligned}$$

$$\text{Again, } \theta_2 = 45^\circ + 15^\circ = 60^\circ$$

$$\begin{aligned}\therefore W_2 &= MB(\cos 45^\circ - \cos 60^\circ) \\ &= MB\left(\frac{1}{\sqrt{2}} - \frac{1}{2}\right) \\ &= MB\left(\frac{\sqrt{2}-1}{2}\right)\end{aligned}$$

$$\begin{aligned}\therefore \frac{W_2}{W_1} &= \frac{MB\left(\frac{\sqrt{2}-1}{2}\right)}{MB\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)} = \frac{1}{\sqrt{2}} \\ \Rightarrow W_2 &= \frac{W_1}{\sqrt{2}} = \frac{W}{\sqrt{2}}\end{aligned}$$

Question4

Materials suitable for permanent magnets should have

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Options:

A.

Low retentivity and low coercivity

B.

Low retentivity and high coercivity

C.

high retentivity and low coercivity

D.

high retentivity and high coercivity

Answer: D

Solution:

Materials suitable for permanent magnets should have high retentivity and high coercivity.

Question5

A short bar magnet of magnetic moment 10^4 JT^{-1} is free to rotate in a horizontal plane. The work done in rotating the magnet slowly from the direction parallel to a horizontal magnetic field of $4 \times 10^{-5} \text{ T}$ to a direction 60° to the direction of the field is

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Options:

A.

0.2 J

B.

2.6 J

C.

0.4 J

D.

6.2 J

Answer: A

Solution:

Given data

- Magnetic moment of the bar magnet:

$$M = 10^4 \text{ J T}^{-1}$$

- Magnetic field:

$$B = 4 \times 10^{-5} \text{ T}$$

- Initial orientation: $\theta_1 = 0^\circ$ (parallel to B)
- Final orientation: $\theta_2 = 60^\circ$

Relevant concept

The **potential energy** of a magnetic dipole in a magnetic field is

$$U = -MB \cos \theta$$

When the magnet is rotated slowly (quasi-statically), the work done **by an external agent** equals the **change in potential energy** of the system:

$$W = U_2 - U_1 = [-MB \cos \theta_2] - [-MB \cos \theta_1]$$

$$\Rightarrow W = MB(\cos \theta_1 - \cos \theta_2)$$

Substitute the values

$$W = (10^4)(4 \times 10^{-5})(\cos 0^\circ - \cos 60^\circ)$$

Compute step by step:

$$\cos 0^\circ = 1, \quad \cos 60^\circ = \frac{1}{2}$$

So,

$$W = (10^4)(4 \times 10^{-5})(1 - 0.5)$$

$$W = (10^4)(4 \times 10^{-5})(0.5)$$

$$W = (10^4 \times 4 \times 10^{-5} \times 0.5)$$

$$W = (4 \times 10^{-1} \times 0.5) = 0.2 \text{ J}$$

 **Final Answer:**

$W = 0.2 \text{ J}$

Correct option: (A) 0.2 J

Question6

A short bar magnet has a magnetic moment of 0.48JT^{-1} . The magnitude of magnetic field at a point at 10 cm distance from the centre of the magnet on its axis is

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Options:

A.

0.96 gauss

B.

0.48 gauss

C.

1.92 gauss

D.

1.44 gauss

Answer: A

Solution:

$$B = \frac{\mu_0}{4\pi} \cdot \frac{2M}{r^3} = 10^{-7} \times \frac{2 \times 0.48}{(0.1)^3}$$

$$= 0.96 \times 10^{-4} \text{ T} = 0.96 \text{ Gauss}$$

Question7

A short bar magnet is placed in a uniform magnetic field of 2 T such that the axis of the magnet makes an angle of 45° with the direction of the magnetic field. If the torque acting on the magnet is $0.36\sqrt{2} \text{ N} - \text{m}$, then the moment of the magnet is

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Options:

A.

$$0.54 \text{ JT}^{-1}$$

B.

$$0.18 \text{ JT}^{-1}$$

C.

$$0.72 \text{ JT}^{-1}$$

D.

$$0.36 \text{ JT}^{-1}$$

Answer: D

Solution:

Torque on bar magnet,

$$\tau = MB \sin \theta$$

$$\Rightarrow M = \frac{\tau}{B \sin \theta} = \frac{0.36\sqrt{2}}{2 \sin 45^\circ}$$

$$= \frac{0.36\sqrt{2}}{2 \times \frac{1}{\sqrt{2}}} = \frac{0.36\sqrt{2}}{\sqrt{2}} = 0.36 \text{ J/T}$$

Question8

The relation between μ and H for a specimen of iron is

$\mu = \left[\frac{1.4}{H} + 12 \times 10^{-4} \right] \text{Hm}^{-1}$. The value of H which produces flux density of 1 T will be (μ = magnetic permeability, H = magnetic intensity)

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Options:

A. 250Am^{-1}

B. 500Am^{-1}

C. 750Am^{-1}

D. 10^3Am^{-1}

Answer: B

Solution:

To find the value of H that produces a flux density of 1 T, we are given:

$$\mu = \left[\frac{0.4}{H} + 12 \times 10^{-4} \right] \text{H/m}$$

We also know the relationship between B and μH :

$$B = \mu H$$

Given $B = 1 \text{ T}$, substitute into the equation:

$$\mu = \frac{0.4}{H} + 12 \times 10^{-4}$$

Multiply both sides by H :

$$\mu H = 0.4 + 12 \times 10^{-4} H$$

Since $B = \mu H = 1 \text{ T}$, substitute the expression for μH :

$$1 = 0.4 + 12 \times 10^{-4} H$$

Rearrange to solve for H :

$$1 - 0.4 = 12 \times 10^{-4} H$$

$$0.6 = 12 \times 10^{-4} H$$

$$H = \frac{0.6}{12 \times 10^{-4}}$$

$$H = \frac{6000}{12}$$

$$H = 500 \text{ A/m}$$

Thus, the value of H is 500 Am^{-1} .

Question9

At a place the horizontal component of earth's magnetic field $3 \times 10^{-5} \text{ T}$ and the magnetic declination is 30° . A compass needle of magnetic moment 18 Am^2 pointing towards geographic north at this place experiences a torque of

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Options:

A. $36 \times 10^{-5} \text{ Nm}$

B. $18 \times 10^{-5} \text{ Nm}$

C. $54 \times 10^{-5} \text{ Nm}$

D. $27 \times 10^{-5} \text{ Nm}$

Answer: D

Solution:

To calculate the torque experienced by a compass needle due to the Earth's magnetic field, we use the following information:

The **angle of declination** is $\alpha = 30^\circ$.

The **magnetic moment** of the compass is $M = 18 \text{ Am}^2$.

The **magnetic field** is $B = 3 \times 10^{-5} \text{ T}$.

The formula to calculate torque (τ) is:

$$\tau = MB \sin \theta$$

where θ is the angle between the magnetic moment and the magnetic field, which in this case is the angle of declination, 30° . Substituting the given values into the formula:

$$\tau = 18 \times 3 \times 10^{-5} \times \sin 30^\circ$$

Since $\sin 30^\circ = \frac{1}{2}$, the calculation becomes:

$$\tau = 18 \times 3 \times 10^{-5} \times \frac{1}{2}$$

$$\tau = 27 \times 10^{-5} \text{ Nm}$$

Therefore, the torque experienced by the compass needle is $27 \times 10^{-5} \text{ Nm}$.

Question10

If the vertical component of the earth's magnetic field is 0.45 G at a location and angle of dip is 60° , then magnetic field of earth in that location is

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Options:

A. 0.26 G

B. 0.52 G

C. 0.3 G

D. 0.7 G

Answer: B

Solution:

Given the vertical component of the Earth's magnetic field as $B_V = 0.45 \text{ G}$ and the angle of dip $\delta = 60^\circ$, we can determine the Earth's magnetic field at this location.

Calculate the Horizontal Component (B_H):

The formula for the angle of dip is:

$$\tan \delta = \frac{B_V}{B_H}$$

Given $\delta = 60^\circ$, we find that:

$$B_H = \frac{B_V}{\tan 60^\circ} = \frac{0.45}{\sqrt{3}} \approx 0.26 \text{ G}$$

Calculate the Earth's Magnetic Field (B_e):

The Earth's total magnetic field can be found using the horizontal component and the angle of dip:

$$\cos \delta = \frac{B_H}{B_e}$$

With $\cos 60^\circ = 0.5$, we have:

$$B_e = \frac{B_H}{\cos 60^\circ}$$

$$= \frac{0.26}{0.5} = 0.52 \text{ G}$$

Thus, the magnetic field of Earth at this location is 0.52 G.

Question11

At a place where the magnitude of the earth's magnetic field is $4 \times 10^{-5} \text{ T}$, a short bar magnet is placed with its axis perpendicular to the earth's magnetic field direction. If the resultant magnetic field at a point at a distance of 40 cm from the centre of the magnet on the normal bisector of the magnet is inclined at 45° with the earth's field, then the magnele moment of the magnet is

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Options:

A. 38.4 Am^2

B. 51.2 Am^2

C. 12.8 Am^2

D. 25.6 Am^3

Answer: D

Solution:

Given:

Earth's magnetic field, $B_e = 4 \times 10^{-5} \text{ T}$

Distance from the magnet, $r = 0.4 \text{ m}$

The magnetic field due to a short bar magnet at a point on its normal bisector is given by:

$$B_m = \frac{\mu_0}{4\pi} \frac{M}{r^3}$$

For a 45° angle of inclination of the resultant magnetic field with Earth's field, we have:

$$B_m = B_e$$

Substituting the values, we get:

$$4 \times 10^{-5} = \frac{\mu_0}{4\pi} \frac{M}{(0.4)^3}$$

This simplifies to:

$$4 \times 10^{-5} = \frac{10^{-7} \times M}{(0.4)^3}$$

Solving for M , the magnetic moment:

$$M = 2.56 \times 10^1$$

Therefore, the magnetic moment of the magnet is:

$$M = 25.6 \text{ Am}^2$$

Question12

Any magnetic material loses its magnetic property when it is

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Options:

- A. dipped in water
- B. dipped in sand
- C. attached to an iron piece
- D. heated to high temperature

Answer: D

Solution:

The correct answer is Option D: heated to high temperature.

Here's why:

Magnetic materials (especially ferromagnetic ones like iron) have a special temperature called the Curie temperature.

When heated above this Curie temperature, the thermal agitation disrupts the alignment of the magnetic domains.

As a result, the material loses its permanent magnetic properties.

Thus, heating the material to a high temperature causes it to lose its magnetism.

Question13

The domain in ferromagnetic material is in the form of a cube of side $2\mu\text{ m}$. Number of atoms in that domain is 9×10^{10} and each atom has a dipole moment of $9 \times 10^{-24}\text{ Am}^2$. The magnetisation of the domain is (approximately).

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Options:

A. $10 \times 10^4\text{ Am}^{-1}$

B. $8 \times 10^4\text{ Am}^{-1}$

C. $12 \times 10^4\text{ Am}^{-1}$

D. $9 \times 10^4\text{ Am}^{-1}$

Answer: D

Solution:

To calculate the magnetization of a ferromagnetic material domain, follow these steps:

Determine the Volume of the Domain:

The domain is a cube with each side measuring $2\mu\text{m}$. First, convert this measurement to meters:

$$2\mu\text{m} = 2 \times 10^{-6}\text{ m}$$

Therefore, the volume V of the cube is:

$$V = (2 \times 10^{-6})^3 = 8 \times 10^{-18}\text{ m}^3$$

Use Given Values:

Number of atoms in the domain, $N = 9 \times 10^{10}$.

Dipole moment of each atom, $M = 9 \times 10^{-24}\text{ Am}^2$.

Calculate the Total Dipole Moment:

The total dipole moment M' for all the atoms is given by:

$$M' = N \times M = 9 \times 10^{10} \times 9 \times 10^{-24} = 81 \times 10^{-14}\text{ Am}^2$$

Find the Magnetization:

Magnetization is the total dipole moment M' divided by the volume V :

$$\text{Magnetization} = \frac{M'}{V} = \frac{81 \times 10^{-14}}{8 \times 10^{-18}} = 9 \times 10^4 \text{ Am}^{-1}$$

Thus, the magnetization of the domain is approximately $9 \times 10^4 \text{ Am}^{-1}$.

Question 14

A magnet suspended in a uniform magnetic field is heated, so as to reduce its magnetic moment by 19%. By doing this, the time period of the magnet approximately

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Options:

- A. increase by 11%
- B. decrease by 19%
- C. increases by 19%
- D. decreases by 4%

Answer: A

Solution:

We know that the time period for an oscillating magnet is given by:

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

where:

M is the magnetic moment,

I is the moment of inertia, and

B_H is the horizontal component of the Earth's magnetic field.

Given that the magnetic moment decreases by 19%, we can express the initial magnetic moment as $M_1 = m$ and the reduced magnetic moment as:

$$M_2 = m - \frac{19 \times m}{100} = 0.81m$$

From the equation for the time period, we derive:

$$\frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}}$$

Substituting the values we have:

$$\frac{T_1}{T_2} = \sqrt{\frac{0.81m}{m}} = \sqrt{0.81} = \frac{9}{10}$$

Rearranging gives:

$$T_2 = \frac{10}{9}T_1$$

This implies:

$$T_2 = 1.11T_1$$

So, the new time period T_2 is $T_1 + 0.11T_1$, which indicates that:

The time period increases by 11%.

Question15

Two short magnets of equal dipole moments M are fastened perpendicularly at their centres. The magnitude of the magnetic field at a distance d from the centre on the bisector of the right angle is (μ_0 = Permeability of free space)

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Options:

A. $\frac{\mu_0}{4\pi} \frac{2\sqrt{2}M}{d^3}$

B. $\frac{\mu_0}{4\pi} \frac{5M}{d^3}$

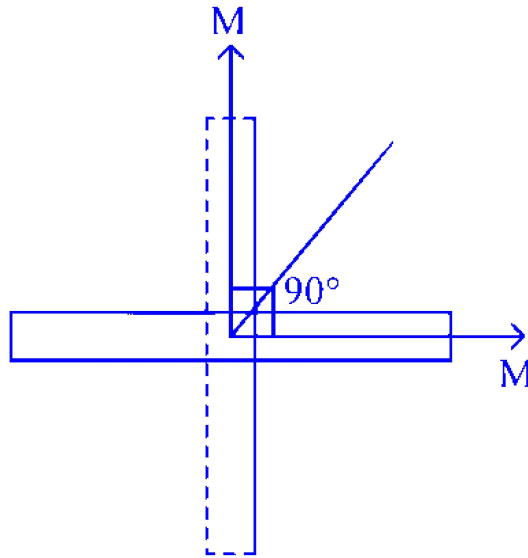
C. $\frac{\mu_0}{4\pi} \frac{2M}{d^3}$

D. $\frac{\mu_0}{4\pi} \frac{10M}{d^3}$

Answer: A

Solution:

When two short magnets of equal dipole moments M are fastened perpendicularly at their centres. Then their resultant dipole moment is given as



$$\begin{aligned}
 M' &= \sqrt{M_1^2 + M_2^2 + 2M_1M_2 \cos \theta} \\
 &= \sqrt{M^2 + M^2 + 2MM \cos 90^\circ} \\
 \Rightarrow M' &= M\sqrt{2} \quad \dots (i)
 \end{aligned}$$

The magnitude of the magnetic field at a distance d from the centre on the bisector i.e at a distance d on the axial position of dipole,

$$\begin{aligned}
 B &= \frac{\mu_0}{4\pi} \cdot \frac{2M'}{d^3} \\
 &= \frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2}M}{d^3} \quad [\text{from Eq. (i)}]
 \end{aligned}$$

Question16

A steel wire of length l and magnetic moment M is bent into a semicircular arc of radius R . The new magnetic moment is

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Options:

- A. M
- B. $\frac{2RM}{\pi l}$
- C. $\frac{2M}{\pi}$
- D. $\frac{2\pi RM}{1}$

Answer: C

Solution:

Length of steel wire = l

Magnetic moment, $M = ml$ (i)

Where, m is pole strength when steel wire is bent into semicircular wire, then

$$l = \pi R$$

$$\Rightarrow R = \frac{l}{\pi}$$

$\therefore$ length of wire = diameter

$$\Rightarrow l' = 2R = \frac{2l}{\pi}$$

$$\Rightarrow l' = \frac{2l}{\pi}$$

New magnetic dipole moment,

$$M' = ml'$$

$$= m \frac{2l}{\pi} = \frac{2}{\pi}(ml) = \frac{2M}{\pi}$$

Question17

A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip pointing down at 30° with the horizontal. The horizontal component of the earth's magnetic field at the place is 0.3 G . Then the magnitude of the earth's magnetic field at the location is

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Options:

A. $\frac{\sqrt{3}}{5}G$

B. $\sqrt{3}G$

C. $\frac{20}{\sqrt{3}}G$

D. $\frac{2}{\sqrt{3}}G$

Answer: A

Solution:

Given:

$$B_H = 0.3G$$

Angle of dip:

$$\delta = 30^\circ$$

We know the relationship between the horizontal component and the total magnetic field:

$$B_H = B \cos \delta$$

Solving for B :

$$\begin{aligned}\Rightarrow B &= \frac{B_H}{\cos \delta} \\ &= \frac{0.3}{\cos 30^\circ} \\ &= \frac{0.3}{\sqrt{3}/2} \\ &= 2 \times \frac{0.3}{\sqrt{3}} \\ &= \frac{0.6}{\sqrt{3}} \\ &= \frac{2}{10} \times \frac{3}{\sqrt{3}} \\ &= \frac{1}{5} \sqrt{3} \\ &= \frac{\sqrt{3}}{5}G\end{aligned}$$

Question18

A compass needle oscillates 20 times per minute at a place where the dip is 45° and the magnetic field is B_1 . The same needle oscillates 30 times per minute at a place where the dip is 30° and magnetic field is B_2 . Then, $B_1 : B_2$ is

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Options:

A. $9\sqrt{3} : 4\sqrt{2}$

B. $4\sqrt{2} : 9\sqrt{3}$

C. $3\sqrt{3} : 2\sqrt{2}$

D. $2\sqrt{2} : 3\sqrt{3}$

Answer: D

Solution:

When compass needle oscillates 20 times per minute, then its time period,

$$T_1 = \frac{1}{f} = \frac{1}{\left(\frac{20}{60}\right)} = 3 \text{ s}$$

Angle of dip, $\delta_1 = 45^\circ$

When the same oscillates 30 times per min then, its time period.

$$T_2 = \frac{1}{f_2} = \frac{1}{30/60} = 2 \text{ s}$$

and angle of dip, $\delta_2 = 30^\circ$

we know that, time period of compass needle

$$T = 2\pi\sqrt{\frac{I}{MB_H}}$$

where, I = moment of inertia B_H = horizontal component of Earth's magnetic field.

$$\Rightarrow T \propto \frac{1}{\sqrt{B_H}}$$

$$\Rightarrow T \propto \frac{1}{\sqrt{B \cos \delta}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{B_1 \cos \delta_1}{B_2 \cos \delta_2}}$$

$$= \sqrt{\frac{B_1 \cos 45^\circ}{B_2 \cos 30^\circ}} = \sqrt{\frac{B_1}{B_2}} = \sqrt{\frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{3}}{2}}} = \sqrt{\frac{2}{\sqrt{6}}} \cdot \sqrt{\frac{B_1}{B_2}}$$

$$\Rightarrow \frac{T_2^2}{T_1^2} = \frac{2}{\sqrt{6}} \cdot \frac{B_1}{B_2} \Rightarrow \frac{2^2}{3^2} = \frac{2}{\sqrt{6}} \frac{B_1}{B_2}$$

$$\Rightarrow \frac{4}{9} = \frac{2}{\sqrt{6}} \frac{B_1}{B_2} \Rightarrow \frac{B_1}{B_2} = \frac{2\sqrt{6}}{9}$$

$$\Rightarrow \frac{B_1}{B_2} = \frac{2\sqrt{2}}{3\sqrt{3}} \Rightarrow B_1 : B_2 = 2\sqrt{2} : 3\sqrt{3}$$

Question19

A paramagnetic sample showing a net magnetisation of 0.8 A m^{-1} , when placed in an external magnetic field of strength 0.8 T , at a temperature 5 K . If the temperature is raised to 20 K , then the magnetisation becomes

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Options:

- A. 0.8 A m^{-1}
- B. 0.2 A m^{-1}
- C. 0.1 A m^{-1}
- D. 0.4 A m^{-1}

Answer: B

Solution:

Given,

Initial magnetisation, $m_1 = 0.8 \text{ Am}^{-1}$

Magnetic field strength, $B = 0.8 \text{ T}$

Initial temperature, $T_1 = 5 \text{ K}$

Final temperature, $T_2 = 20 \text{ K}$

Let, final magnetisation be m_2 .

Since, $m = CB/T$

where, C is Curie's temperature.

$$\begin{aligned}\Rightarrow m &\propto \frac{1}{T} \\ \therefore \frac{m_1}{m_2} &= \frac{T_2}{T_1} \\ \Rightarrow m_2 &= \frac{m_1 T_1}{T_2} = \frac{0.8 \times 5}{20} = 0.2 \text{ Am}^{-1}\end{aligned}$$

Question20

The plane of a dip circle is set in the geographic meridian and the apparent dip is δ_1 . It is then set in a vertical plane perpendicular to the geographic meridian. The apparent dip angle is δ_2 . The declination θ at the place is

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Options:

- A. $\tan^{-1}(\tan_1 \tan_2)$
- B. $\tan^{-1}(\tan \delta_1 + \tan \delta_2)$
- C. $\tan^{-1}\left(\frac{\tan \delta_1}{\tan \delta_2}\right)$
- D. $\tan^{-1}(\tan \delta_1 - \tan \delta_2)$

Answer: C

Solution:

Given, first apparent dip = δ_1

Second apparent dip = δ_2

Declination = θ

As we know that, $\tan \theta = \frac{\tan_1}{\tan_2}$

$$\therefore \theta = \tan^{-1}\left(\frac{\tan \delta_1}{\tan \delta_2}\right)$$
